INTERNAL BENCHMARK SUMMARY — CERTIVA HYBRID SEARCH ENGINE

Arsiva Tech · Internal audit document (not a marketing claim)

Run 2026-10-14 00:20:43 | Commit `f5c1ba1` (MCP v3.3) · original snapshot: September 11, 2026 (commit `32fa992`)

Headline Metrics

Search latency	`7.1 ms — official 103-query run, Oct 14, 2026`
Precision@1 (75 labelled scenarios)	`73/75 (97.3%)`
Recall@5 (75 labelled scenarios)	`74/75 (98.7%)`
Guardrail pass (28 out-of-domain queries, threshold 0.35)	`28/28 (100%)`

Honest Notes (read before using these numbers)

1. One labelled scenario (semantic path, empty top-1) remains FAIL — a known limitation signed off on Sept 11, 2026 (Option 1 chosen; correction: official Jakarta Finals date Oct 13, 2026, finalist selection Sept 22). Decision: deferred after the finals, not hidden.
2. Guardrail: the Sept 11 snapshot recorded 27/28 (1 residual leak, the "OTG for phone data transfer" query). The official Sept 19, 2026 run (commit `2ca12e7`) and the Oct 9, 2026 re-run (commit `f5c1ba1`): 28/28 (100%) — gap closed, confirmed twice.
3. P@1 73/75 includes the acronym query 'PPN': top-1 = PMK 131/2024, expected label = UU 8/1983. Both documents are valid answers to that query — a label ambiguity, not a hallucination.
4. The numbers above are a snapshot (not a live feed). ±1 query fluctuation between full runs is normal; official reports are always stamped with run_at + commit.

How to Reproduce

1. `cd C:/certiva/acse-cocreate-jakarta`
2. `python -m venv .venv` then activate (Windows: `.venv\Scripts\activate`)
3. `pip install -r requirements.txt` (ONNX runtime; torch not required)
4. `python scripts/benchmark_hybrid.py` → `benchmark_report.md` (stamped run_at + commit)

Detailed Table — 103 Queries (results of the Oct 14, 2026 run)

Query	Type	Expected	Top-1	Via	Status
vendor mancanegara lapak	regression	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
apa saja syarat menjadi importir digital atau pedagang luar negeri?	regression	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
kewajiban notifikasi kebocoran data	regression	UU-27-2022, PP-71-2019	UU-27-2022	bm25+semantic	PASS
pajak marketplace	regression	PMK-37-2025	PMK-37-2025	bm25+semantic	PASS
denda data breach	regression	UU-27-2022, PP-71-2019	UU-27-2022	bm25+semantic	PASS
pinjaman online laporan harian	regression	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
keamanan sistem elektronik	regression	PP-71-2019, PERMENDAG-19-2026, UU-1-2024, PMK-155-2022	PP-71-2019	bm25+semantic	PASS
identitas digital sertifikasi elektronik	regression	UU-1-2024	UU-1-2024	bm25+semantic	PASS
kewajiban PEB ekspor	regression	PMK-155-2022, PMK-4-2025	PMK-155-2022	bm25+semantic	PASS
fiksi hukum barang dimuat sarana pengangkut	regression	UU-17-2006, PMK-155-2022	UU-17-2006	bm25+semantic	PASS
dasar sah pemrosesan data pribadi	regression	UU-27-2022, PP-71-2019	UU-27-2022	bm25+semantic	PASS
sanksi administratif pelanggaran data pribadi	regression	UU-27-2022, PP-71-2019	UU-27-2022	bm25+semantic	PASS
kontrak elektronik internasional	regression	UU-1-2024	UU-1-2024	bm25+semantic	PASS
perlindungan anak penyelenggara sistem elektronik	regression	UU-1-2024, PP-71-2019, PERMENDAG-19-2026, PMK-155-2022	UU-1-2024	bm25+semantic	PASS
kewajiban pendaftaran penyelenggara sistem elektronik	regression	PP-71-2019, PERMENDAG-19-2026, UU-1-2024, PMK-155-2022, PMK-4-2025	PP-71-2019	bm25+semantic	PASS
pemutusan akses sistem elektronik melanggar	regression	PP-71-2019, PERMENDAG-19-2026, UU-1-2024, PMK-155-2022	PP-71-2019	bm25+semantic	PASS
perwakilan PPMSE luar negeri	regression	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
bukti pemungutan PPh 22 marketplace	regression	PMK-37-2025	PMK-37-2025	bm25+semantic	PASS
audit internal sistem pelaporan fintech	regression	POJK-8-2026, PP-71-2019	POJK-8-2026	bm25+semantic	PASS
larangan jual beli data pengguna pindar	regression	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
batas berat paket ekspor pakai consignment note	regression	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS
cara kirim barang UMKM ke luar negeri lewat kurir atau pos	disambiguation	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS
pengecualian larangan pembatasan ekspor kiriman pribadi bukan usaha	regression	PMK-4-2025, PERMENDAG-16-2025, PMK-155-2022	PMK-4-2025	bm25+semantic	PASS
konsolidasi PKBK penyelenggara pos ekspor	regression	PMK-4-2025, PMK-155-2022	PMK-4-2025	bm25+semantic	PASS
rekonsiliasi outward manifest barang kiriman	disambiguation	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS
syarat importir	regression	PERMENDAG-16-2025	PERMENDAG-16-2025	bm25+semantic	PASS
syarat jadi importir apa saja	regression	PERMENDAG-16-2025	PERMENDAG-16-2025	bm25+semantic	PASS
perbedaan API-U dan API-P	regression	PERMENDAG-16-2025, PP-29-2021	PERMENDAG-16-2025	bm25+semantic	PASS
dasar hukum NIB berlaku sebagai angka pengenal importir	disambiguation	PP-29-2021, PERMENDAG-16-2025, UU-17-2006, UU-27-2022, UU-8-1983	PP-29-2021	bm25+semantic	PASS
boleh gak importir pegang API-U dan API-P sekaligus	disambiguation	PERMENDAG-16-2025, PP-29-2021	PERMENDAG-16-2025	bm25+semantic	PASS
barang yang dilarang diperdagangkan pemilik API-P	regression	PERMENDAG-16-2025, PP-29-2021	PERMENDAG-16-2025	bm25+semantic	PASS
berapa persen PPN barang impor	regression	PMK-131-2024, UU-8-1983, PERMENDAG-16-2025, PP-29-2021, UU-17-2006	PMK-131-2024	bm25+semantic	PASS
PPN ekspor 0 persen dasar hukumnya apa	disambiguation	UU-8-1983, PMK-155-2022, PMK-131-2024	UU-8-1983	bm25+semantic	PASS
cara hitung PPN barang bukan mewah nilai lain	regression	PMK-131-2024, UU-8-1983, PMK-4-2025, PERMENDAG-16-2025	PMK-131-2024	bm25+semantic	PASS
tarif PPN naik jadi berapa persen	disambiguation	UU-8-1983, PMK-131-2024	UU-8-1983	bm25+semantic	PASS
PPN barang mewah kendaraan bermotor berapa persen	regression	PMK-131-2024, UU-8-1983	PMK-131-2024	bm25+semantic	PASS
apa saja yang kena objek pajak pertambahan nilai	regression	UU-8-1983, PMK-131-2024	UU-8-1983	bm25+semantic	PASS
kalau aplikasi startup saya tiba-tiba error dan pengguna komplain, siapa yang bisa dituntut secara hukum?	semantic_recovery	(kosong)	(kosong)	-	**FAIL** ■ P@1-MISS
resep sambal terasi paling enak	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
jadwal pertandingan bulutangkis minggu ini	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
cara merawat kucing anggora	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
review film horor terbaru	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
tips belajar main gitar untuk pemula	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
PEB	regression_akronim	PMK-155-2022	PMK-155-2022	bm25+semantic	PASS
PKBK	regression_akronim	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS

Query	Type	Expected	Top-1	Via	Status
DPO	regression_akronim	UU-27-2022	UU-27-2022	bm25+semantic	PASS
Pusdafil	regression_akronim	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
LPBBTI	regression_akronim	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
OJK	regression_akronim	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
FDC	regression_akronim	POJK-8-2026	POJK-8-2026	bm25+semantic	PASS
dan	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
yang	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
untuk	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
di	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
dari	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
atau	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
tidak	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
dengan	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
jadwal kereta api jakarta surabaya	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
tips diet sehat menurunkan berat badan	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
cara menanam cabai di pot	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
cara main catur untuk pemula	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
cara memperbaiki wifi yang lambat	guardrail_nonsense	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
PSE	must_pass_akronim	PP-71-2019, UU-1-2024	PP-71-2019	bm25+semantic	PASS
PDP	must_pass_akronim	UU-27-2022, PP-71-2019	UU-27-2022	bm25+semantic	PASS
NIB	must_pass_akronim	PP-29-2021, PERMENDAG-16-2025, PERMENDAG-19-2026	PP-29-2021	bm25+semantic	PASS
PPN	must_pass_akronim	PMK-131-2024, UU-8-1983	PMK-131-2024	bm25+semantic	PASS ■ P@1-MISS
PMSE	must_pass_akronim	PERMENDAG-19-2026, PMK-37-2025	PERMENDAG-19-2026	bm25+semantic	PASS
PPMSE	must_pass_akronim	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
ITE	must_pass_akronim	UU-1-2024, PP-71-2019	UU-1-2024	bm25+semantic	PASS
OSS	must_pass_akronim	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
PSTE	must_pass_akronim	PP-71-2019	PP-71-2019	bm25+semantic	PASS
DJBC	must_pass_akronim	UU-17-2006	UU-17-2006	bm25	PASS
SKP	must_pass_akronim	PMK-155-2022	PMK-155-2022	bm25+semantic	PASS
CN	must_pass_akronim	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS
UMKM	must_pass_akronim	PMK-4-2025, PMK-37-2025, PMK-131-2024	PMK-4-2025	bm25+semantic	PASS
API	must_pass_akronim	PERMENDAG-16-2025, PP-29-2021	PERMENDAG-16-2025	bm25+semantic	PASS
tarif PPN impor	must_pass_frasa	PMK-131-2024, UU-8-1983, PMK-4-2025	PMK-131-2024	bm25+semantic	PASS
NIB API	must_pass_frasa	PP-29-2021, PERMENDAG-16-2025	PP-29-2021	bm25+semantic	PASS
PEB BC 3.0	must_pass_frasa	PMK-155-2022	PMK-155-2022	bm25+semantic	PASS
sertifikat kompetensi pedagang	must_pass_frasa	PERMENDAG-19-2026	PERMENDAG-19-2026	bm25+semantic	PASS
denda 2 miliar PSE	must_pass_frasa	PP-71-2019	PP-71-2019	bm25+semantic	PASS
de minimis USD 3	must_pass_frasa	PMK-4-2025	PMK-4-2025	bm25+semantic	PASS
neraca komoditas impor	must_pass_frasa	PP-29-2021, PMK-4-2025	PP-29-2021	bm25+semantic	PASS
PPh 22 Pihak Lain	must_pass_frasa	PMK-37-2025, POJK-8-2026	PMK-37-2025	bm25+semantic	PASS
DPO data sensitif	must_pass_frasa	UU-27-2022	UU-27-2022	bm25+semantic	PASS
audit keamanan sistem elektronik	must_pass_frasa	PP-71-2019, PERMENDAG-19-2026, UU-1-2024, PMK-155-2022	PP-71-2019	bm25+semantic	PASS
PMK-4-2025	must_pass_regulasi	PMK-4-2025, PMK-37-2025, UU-8-1983, PMK-131-2024	PMK-4-2025	bm25_exact_lookup	PASS
POJK-8-2026	must_pass_regulasi	POJK-8-2026	POJK-8-2026	bm25_exact_lookup	PASS
PERMENDAG-19-2026	must_pass_regulasi	PERMENDAG-19-2026, PERMENDAG-16-2025	PERMENDAG-19-2026	bm25_exact_lookup	PASS
UU-27-2022	must_pass_regulasi	UU-27-2022, UU-1-2024, PMK-155-2022, UU-8-1983	UU-27-2022	bm25_exact_lookup	PASS
PMK-155-2022	must_pass_regulasi	PMK-155-2022	PMK-155-2022	bm25_exact_lookup	PASS
PP-29-2021	must_pass_regulasi	PP-29-2021, PERMENDAG-16-2025	PP-29-2021	bm25_exact_lookup	PASS
SKCK online berapa hari selesai	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
kapan piala dunia 2026 dimulai	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail

Query	Type	Expected	Top-1	Via	Status
					aktif)
OTG buat transfer data hp	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
PIN ATM kelupaan lagi	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
jam tangan pria	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
lowongan kerja terdekat	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
info program KUR	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
harga emas hari ini	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
warna cat rumah	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)
jaga pos ronda	guardrail_holdout_probe4	(kosong)	(kosong)	-	PASS (kosong, guardrail aktif)

Revision History

Date	Change
Oct 14, 2026	Re-run with no code change (commit `f5c1ba1`; engine report commit `7287fe8`). P@1 73/75 & R@5 74/75 & guardrail 28/28 re-confirmed — identical numbers; latency 7.1 ms, hybrid mode. MCP v3.3.
Oct 9, 2026	Guardrail 28/28 (100%) — OTG gap closed (confirmed since the Sept 19, 2026 run 2ca12e7, re-validated by the official re-run). P@1 73/75 & R@5 74/75 re-confirmed; latency 6.7 ms. MCP v3.3 (commit f5c1ba1).
Sept 11, 2026	OTG guardrail leak fixed (26/28 → 27/28). Two P@1 issues documented: 1 dense-only miss (semantic path, empty top-1) + 1 acronym ambiguity (PPN). Claim corrected 103/103 → 102/103 (independent audit F16).

Methodology (How These Numbers Were Computed)

1. Queries: 75 real-world scenarios (13 regulations) + 28 guardrails (25 in-sample + 3 holdout). All hand-written; no synthetic queries.
2. Metrics: Precision@1 (top-1 = expected document) and Recall@5 (expected document in top 5). Per-scenario label = single best document (no multi-label).
3. Ranking: BM25 (SQLite FTS5) + local ONNX semantic (paraphrase-multilingual-MiniLM-L12-v2, 384-dim) fused via RRF. No LLM; no API; no external calls.
4. Corpus: 13 core regulations, manually verified against official sources (JDIH BPK / Ministry of Finance) by a non-commercial legal practitioner network (10 notary offices + 1 law firm). 54 articles & 41 relations computed live from the database via corpus_stats.

Integrity Policy

Every number on this site comes from corpus_stats and the 103-query benchmark report (benchmark_report.md, stamped run_at + commit), not static claims. When a number changes, this document is revised with a revision-history entry — no silent changes.

Disclaimer

This document is an internal audit of Certiva (Arsiva Technology). It is not a guarantee of legal advice, and not a certification by any authority. Figures refer to the benchmark snapshot dated above; full re-runs may fluctuate ±1 query.